

A Minor Project Synopsis

On

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Agricultural Pest Detection Using YOLOv8 Based Deep Learning Model

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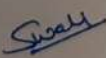
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1. Introduction

Agriculture plays a crucial role in our country's economy and food supply. However, one of the major challenges faced by farmers is crop damage caused by pests. If pests are not identified at the right time, they can spread quickly and cause significant loss in crop yield and quality.

Traditionally, pest detection is done through manual inspection of crops. This method requires time, effort, and experience, and it may not always provide accurate results, especially in large agricultural fields. With advancements in artificial intelligence and computer vision, automated systems can now be developed to detect pests using image-based techniques.

This project focuses on building an automated agricultural pest detection system using the YOLOv8 deep learning model. The system is designed to detect five common agricultural pests: grasshopper, caterpillar, snail, weevil, and slug. The aim is to develop a reliable and efficient solution that can help in early pest detection and support modern farming practices.

2. Scope of the Project

The scope of this project is limited to detecting five specific agricultural pests using image-based deep learning techniques. The project includes collecting pest images, annotating them using bounding boxes, training the YOLOv8 model, evaluating its performance, and testing it on new images.

The system works only with static images and focuses strictly on pest identification. It does not include pesticide recommendations, crop disease detection, or pest population monitoring. The intended users of this system are farmers, agricultural researchers, and developers working in smart agriculture technologies.

Although the current system is limited to image-based detection, it can be extended in the future to support real-time video detection or integration with drones and mobile applications.

3. Problem Statement

Farmers often find it difficult to detect pest infestations at an early stage. Manual monitoring of crops can be time-consuming and may not be practical for large farms. Late detection of pests can result in severe crop damage and financial loss.

The problem addressed in this project is to develop an automated system that can accurately detect agricultural pests from images using deep learning techniques. The system should provide reliable detection results and perform effectively on unseen data.

4. Objectives

The main objectives of this project are:

- To collect and prepare a dataset of agricultural pest images.
- To annotate the images using bounding boxes for object detection.
- To train a YOLOv8 model for detecting multiple pest classes.
- To evaluate the model using performance metrics such as precision, recall, and mean Average Precision (mAP).
- To test the trained model on unseen images to verify its accuracy.
- To demonstrate the practical application of artificial intelligence in agriculture.

5. Literature Review

Agricultural pest detection has evolved significantly over the years. Earlier approaches mainly relied on traditional image processing techniques. These methods extracted features such as color, texture, and shape from images and used machine learning algorithms like Support Vector Machines (SVM) and k-Nearest Neighbors (KNN) for classification. However, these approaches were sensitive to environmental conditions and often failed in complex backgrounds.

With the advancement of deep learning, Convolutional Neural Networks greatly improved object detection accuracy. Models such as Faster R-CNN and SSD were introduced and provided better results compared to traditional methods. Later, the YOLO (You Only Look Once) algorithm became popular because of its speed and real-time detection capability.

Different versions of YOLO, including YOLOv3 and YOLOv5, have been successfully used in agricultural monitoring systems. YOLOv8 further improves detection accuracy and efficiency through architectural enhancements and optimized training techniques. This project builds upon existing research by implementing YOLOv8 for multi-class pest detection and evaluating its performance using standard metrics.

6. Methodology

The project follows a systematic and structured approach.

First, a dataset of 400 images containing five pest categories was collected. The images were carefully reviewed, and duplicate or low-quality images were removed to improve training quality.

Next, the images were annotated using Roboflow by drawing bounding boxes around each pest. The dataset was divided into training, validation, and testing sets in a 70:20:10 ratio.

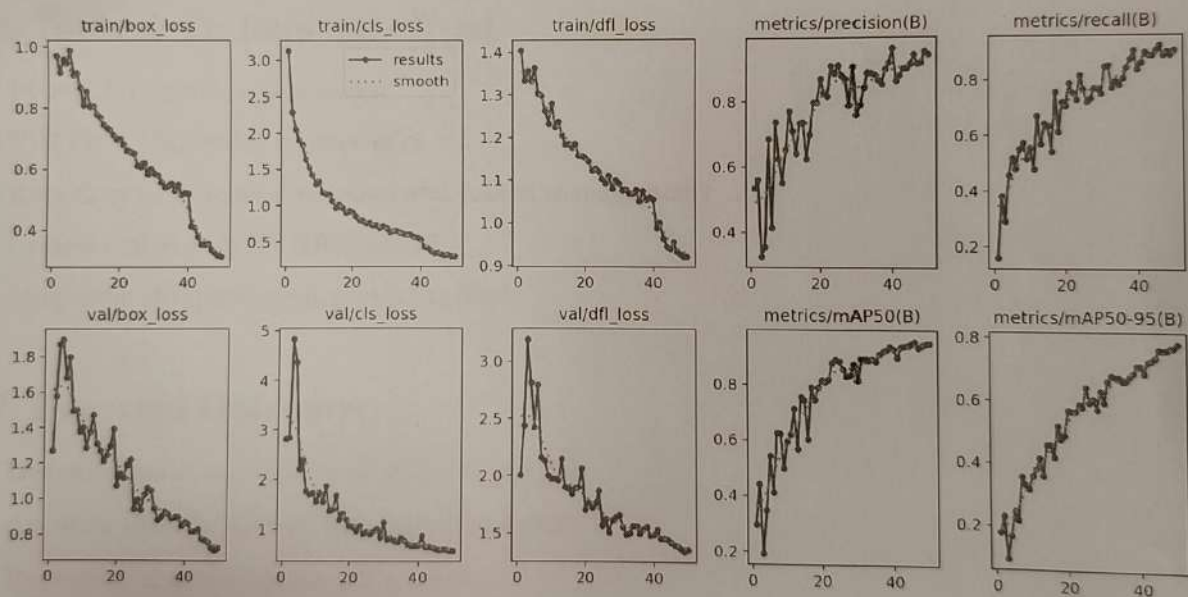
All images were resized to 640×640 pixels, and data augmentation techniques such as horizontal flipping and brightness adjustment were applied to improve the model's generalization ability.

The YOLOv8n model was trained for 50 epochs using Google Colab with NVIDIA T4 GPU support.

During training, performance metrics such as precision, recall, and mAP were monitored.

The model achieved approximately 94% mAP@50 and 80% mAP@50–95. Finally, the model was tested on unseen images to evaluate its real-world performance.

Figure 1 shows the training and validation performance of the model.



7. Use of Engineering

This project demonstrates the practical application of engineering knowledge in solving an agricultural problem.

The system was developed using Python programming and the Ultralytics YOLO framework. Proper development steps were followed, including dataset preparation, preprocessing, model training, testing, and evaluation.

Machine learning concepts such as neural network optimization and GPU acceleration were applied to improve efficiency. Performance evaluation was conducted using engineering metrics such as precision, recall, confusion matrix, and mean Average Precision.

The project shows how theoretical concepts of artificial intelligence can be converted into a practical working system.

8. Tools & Technologies Used

- Python Programming Language
- YOLOv8 (Ultralytics Framework)
- Roboflow (for image annotation and dataset management)
- Google Colab with NVIDIA T4 GPU
- Matplotlib (for performance visualization)

9. Expected Outcomes

- Development of a reliable pest detection model.
- Accurate identification of five agricultural pests.
- Reduction in manual monitoring effort.
- Foundation for future smart farming applications.

10. Applications

- Smart agriculture monitoring systems
- Drone-based crop surveillance
- Mobile pest detection applications
- Precision farming technologies

11. Conclusion

This project demonstrates how deep learning can be effectively applied in agriculture. By implementing the YOLOv8 model, an accurate pest detection system has been developed and validated. The model achieved strong performance results and successfully detected pests in unseen images.

The system can be further improved and expanded for real-time field applications in the future. Overall, the project highlights the importance of artificial intelligence in solving real-world agricultural challenges.